a must be 0. What can we say about the common ratio r? If we take r as 1, 2 or any other number we get the same sequence $0, 0, 0, 0, \ldots$. We are left with the situation where a geometric sequence has infinitely many common ratios. To overcome these confusions some mathematicians exclude this sequence from the class of geometric sequences by assuming $a \neq 0$ in the definition. (We made this assumption)

5.4.2 Arithmetico-Geometric Progression (AGP)

Combining arithmetic and geometric progressions, a new progression called arithmetico geometric progression is formed. As we use the abbreviations AP and GP for arithmetic progressions and geometric progressions, we use the abbreviation AGP for arithmetico geometric progression. AGP's arise in various applications, such as the computation of expected value in probability theory.

Definition 5.1

A sequence of the form

$$a,\ (a+d)r,\ (a+2d)r^2,\ (a+3d)r^3,\ \ldots,\ (a+(n-1)d)r^{n-1},\ (a+nd)r^n,\ \ldots$$

is called an ***arithmetico-geometric progression*** or an ***arithmetico-geometric sequence***.

Consider an AP: $a, a+d, a+2d, \ldots$
GP: $1, r, r^2, \ldots$

Then the AGP is $a, (a+d)r, (a+2d)r^2, \ldots$

Here, a is the ***initial term***, d is the ***common difference*** and r is the ***common ratio*** of the AGP.

If we take $r = 1$, then the AGP will become an AP and if we take $d = 0$, then it will become a GP. So the arithmetic and Geometric progressions become particular cases of AGP. This is a nice situation to know the concept of generalization in mathematics.

We note that the nth term of an AGP is given by $T_n = (a + (n-1)d)r^{n-1}$. All APs and all GPs are AGPs.

For example, the AP $0, 1, 2, 3, 4, \ldots$ and the GP $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$ give the AGP $\frac{0}{1}, \frac{1}{2}, \frac{2}{4}, \frac{3}{8}, \ldots$

The sequence $4, 14, 40, 104, 256, 608, \ldots$ is also an example of an arithmetico-geometric progression. For this sequence $a = 4$, $d = 3$ and $r = 2$.

5.4.3 Harmonic Progression (HP)

Harmonic progression is one of many important sequences and is closely related to the arithmetic progression. Harmonic progression is widely used.

Definition 5.2

A sequence $h_1, h_2, h_3, \ldots$ is said to a ***harmonic sequence*** or a ***harmonic progression*** if $\frac{1}{h_1}, \frac{1}{h_2}, \frac{1}{h_3}, \frac{1}{h_4}, \ldots$ is an arithmetic sequence.

Note that a sequence is in harmonic progression if reciprocal of its term are in arithmetic progression. But we should not say that harmonic progressions are reciprocals of arithmetic progressions; in fact, if an arithmetic sequence contains a zero term, then its reciprocal is not meaningful. Of course, if an arithmetic progression contains no zero term, then its reciprocal is a harmonic progression. So a general harmonic progression will be of the form

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \frac{1}{a+3d}, \ldots$$